

Features to Implement in the Next Deliverable (Deliverable 2)

1. Backend Integration (Node.js)

The system will be connected to a backend server implemented using Node.js. This backend will handle data processing, communication between the client and the database, and the application logic.

Main objectives:

- Create API endpoints for courses and assessments.
- Allow communication between the frontend dashboard and backend services.
- Handle requests and errors properly.

2. Database and Data Persistence

In Deliverable 1, most data was hardcoded for demonstration purposes.

In Deliverable 2, all data will be stored in a database.

The system will store:

- User accounts
- Courses (code, name, instructor, term)
- Assessments (type, due date, grade, weight, status)
- Student progress and grades

Data will be retrieved dynamically when users log in.

3. User Authentication System

A secure authentication system will be implemented so users can access the platform safely.

Features include:

- User login and logout
- Account creation
- Authentication validation for protected pages
- Session management to maintain login state.

4. Role-Based Access Control

Different users will have different permissions.

Students will be able to:

- View courses they are enrolled in
- View assessment deadlines
- View grades and progress

Instructors/Admins will be able to:

- Create and manage courses
- Define assessment categories and weights
- Add or modify assessments
- Enable or disable courses.

5. Server-Side Grade Calculations

In Deliverable 1, grade calculations are done on the frontend.

In Deliverable 2, these calculations will be handled on the server side.

Calculations will include:

- Weighted course averages
- Progress percentages
- Assessment completion tracking.

6. Dynamic Calendar Integration

The current calendar displays hardcoded assessment dates.

In Deliverable 2:

- Assessment due dates will be retrieved from the backend database.
- The calendar will automatically update when new assessments are added.
- Clicking a calendar day will display assessments due on that date.

7. Data Validation

All user inputs will be validated to ensure correct data entry.

Validation will include:

- Required fields
- Valid grade ranges

- Correct date formats
- Prevention of invalid weights.

8. System Architecture Improvements

The project structure will be improved to support scalability.

This includes:

- Separation of frontend and backend code
- Organized API routes
- Clear modular project structure.

9. Optional Bonus Features

If time permits, the following features may be added:

- GPA estimation across courses
- Export grades to PDF or CSV
- Email reminders for upcoming deadlines
- Dark/light mode support